



Transcutaneous Electrical Acupoints Stimulation Improves Spontaneous Voiding Recovery After Laparoscopic Cholecystectomy: A Randomized Clinical Trial

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Abstract

Background Facilitating the recurrence of spontaneous voiding is considered to be a way to prevent urinary retention after surgery, which is of great importance in cholecystectomy. This study aimed to assess the effect of transcutaneous electrical acupoint stimulation (TEAS) on spontaneous voiding recovery after laparoscopic cholecystectomy.

Methods Participants who underwent elective laparoscopic cholecystectomy were randomly assigned to either the TEAS group or the sham group. Active TEAS or sham TEAS at specific acupuncture points was conducted intraoperatively and postoperatively. The primary outcome was the recovery speed of spontaneous voiding ability after surgery and secondary outcomes included postoperative urinary retention (POUR), voiding dysfunction, pain, anxiety and depression, and early recovery after surgery.

Results A total of 1,948 participants were recruited and randomized to TEAS ($n = 975$) or sham ($n = 973$) between August 2018 and June 2020. TEAS shortens the time delay of the first spontaneous voiding after laparoscopic cholecystectomy (5.6 h [IQR, 3.7–8.1 h] in the TEAS group vs 7.0 h [IQR, 4.7–9.7 h] in the sham group) ($p < 0.001$). The TEAS group experienced less POUR ($p = 0.020$), less voiding difficulty ($p < 0.001$), less anxiety and depression ($p < 0.001$), reduced pain ($p = 0.007$), and earlier ambulation ($p = 0.01$) than the sham group.

Conclusions Our results showed that TEAS is an effective approach to accelerate the recovery of spontaneous voiding and reduce POUR which facilitates recovery for patients after laparoscopic cholecystectomy.

Introduction

Gallstone disease is one of the most common and costly gastroenterological disorders. It is estimated that nearly 700,000 cholecystectomy surgeries are performed annually in the USA [1] and the incidence is increasing in developing countries [2, 3]. Urinary retention is one of the major complications after laparoscopic cholecystectomy, affecting 4.7–15.8% patients, and it is associated with prolonged hospital stay, increased costs, and patient anxiety and discomfort [4–6].

Elective laparoscopic cholecystectomy without catheterization has become popular in recent years because of its minimal invasiveness [7]. However, it may impair spontaneous voiding ability due to diminished awareness of bladder sensation after surgical procedures [8]. Delayed

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spontaneous voiding can result in bladder distension, urinary retention, psychological stress, abdominal pain, confusion, and restlessness [9]. Various treatments have been attempted to improve voiding problems, such as alpha blockers, botulin toxin, and bladder catheterization; however, their effects are still controversial [10, 11].

Acupuncture has gained popularity as a non-pharmaceutical treatment for recovering the voiding control function. Recent clinical trials have shown that acupuncture and related methods can reduce the incidence of postoperative urinary retention (POUR) after orthopedic, obstetric, gynecological, and gastrointestinal surgeries [12–14]. Transcutaneous electrical acupoint stimulation (TEAS) applies transcutaneous electrical stimulation to acupuncture points instead of needles and has demonstrated benefits in regulating voiding in animal studies [15, 16]. However, these studies had limitations, including a small sample size, inadequate allocation concealment, etc.

Therefore, we conducted a large-scale, single-center, double-blind randomized trial to assess the efficacy of TEAS for spontaneous voiding recovery after laparoscopic cholecystectomy.

Methods

Study design

This prospective, double-blind trial was conducted at Nankai Hospital, a tertiary hospital trusted by China National Natural Science Foundation between August 21, 2018, and June 17, 2020. This study was approved by the local institutional review board, and written informed consent was obtained from all subjects participating in the trial. The trial was registered prior to patient enrollment at clinicaltrials.gov (NCT03631160) and conducted in accordance with the Declaration of Helsinki.

Patients

Patients were included if they were 40–75 years old, the age at which gallbladder stones are most likely to occur [17]. Exclusion criteria were preventive or intraoperative indwelling catheter, urinary tract obstruction, or infections within 3 months (such as moderate or severe benign prostatic hyperplasia, stress urinary incontinence, urinary tract infections, etc.), critical condition (preoperative American Society of Anesthesiologists, [ASA] grade $\geq$ IV), severe renal impairment (need to undergo renal replacement therapy), severe liver and kidney dysfunction (Child–Pugh grade C), contraindications to TEAS (such as skin allergy, infection, pruritus, or wearing a pacemaker),

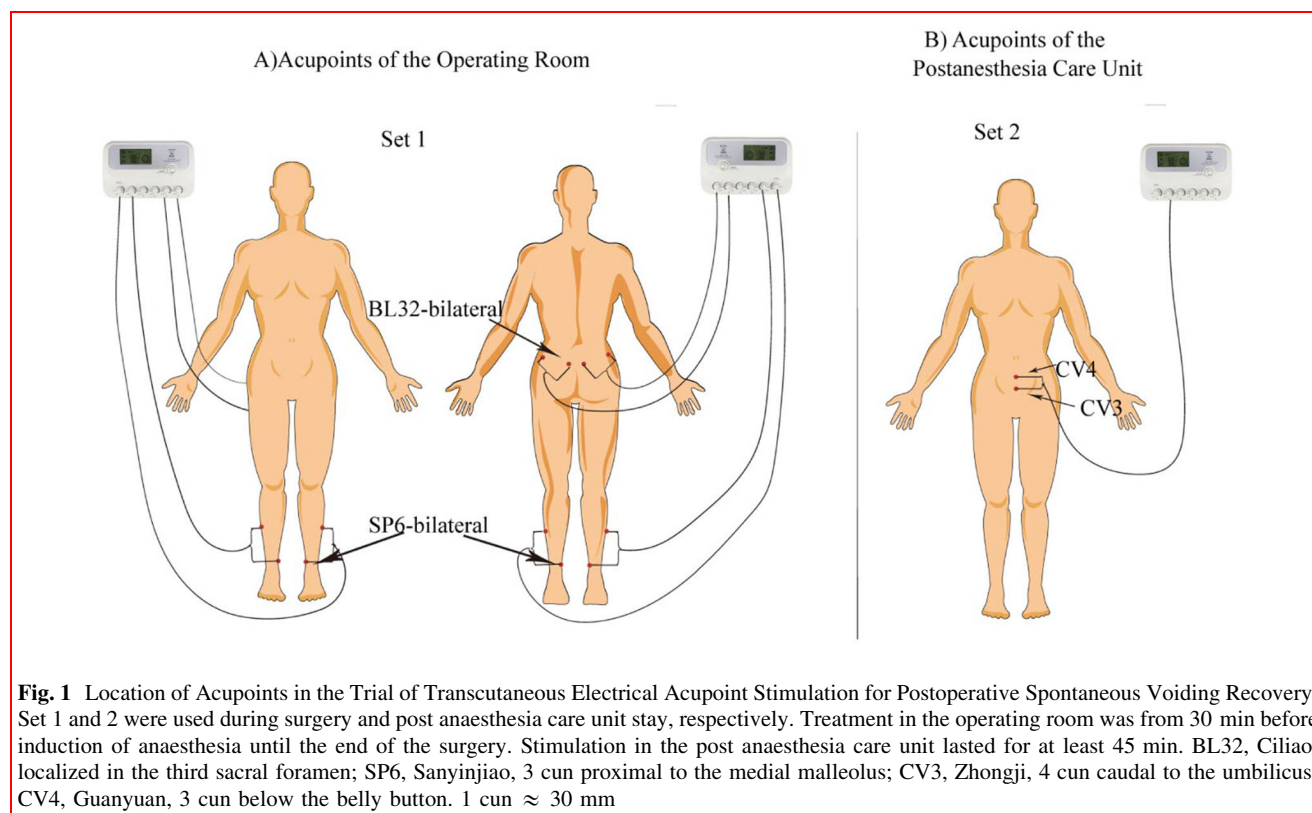
and participating in other clinical investigations within 3 months prior to enrollment in this study. Besides, patients with previous acupuncture treatment experience were excluded for they had a higher probability of acupuncture insensitivity.

Randomization and blinding

Eligible participants were randomly assigned to the TEAS or sham group in a 1:1 ratio using block randomization. The randomized numbers were enclosed in sequentially numbered, sealed, opaque envelopes and revealed by the acupuncturist in charge prior to the start of surgery. The participants, outcome assessors, staff responsible for the postoperative care and statisticians were blinded to the treatment allocation.

Interventions

A full-time acupuncturist who is a registered TCM practitioner with eight years of clinical experience provided TEAS and sham treatments independently. In the intervention, she remained silent regarding the allocation and effects of TEAS. However, she informed every patient before performing the intervention that they may or may not feel a tingling sensation around the calf and lower back during TEAS treatment. The participants in the TEAS group received stimulation at four locations: Ciliao (BL32, located at the third sacral foramen), Sanyinjiao (SP6, located 3 cun proximal to the medial malleolus), Zhongji (CV3, located 4 cun below the belly button), and Guanyuan (CV4, located 3 cun below the belly button) (Fig. 1). In the TEAS group, a low-frequency pulse electroacupuncture stimulator (Hwato brand model no. SDZ—IV, Suzhou Medical Appliances Co., Ltd, Suzhou, China) was attached to four acupuncture points. TEAS was started 30 min before the induction of anesthesia and lasted until 45 min after the surgery. The stimulator applied a 2-Hz sparse wave, and the intensity was gradually increased until the strongest needle-like tingling sensation that patients could tolerate was reached [18]. Participants in the sham group received a sham stimulus at the acupoints by cutting off the output wires and no current output was delivered to the patient, even though it was turned on. This approach is based on a clinical study of acupuncture treatment and shown to be free of de qi reactions in patients [19]. In order to make the simulation as realistic as possible, when switched on, the apparatus with sham electrode lines had the same working power indicator and sound as those with normal electrode lines, suggesting to participants that the apparatus is functional even though it actually had no current output.



All patients voided completely immediately before entering the operating room. Intraoperative monitoring included electrocardiography, noninvasive blood pressure, peripheral capillary oxygen saturation, end-breathing carbon dioxide partial pressure, and bispectral index. Anaesthesia was induced with etomidate (0.2–0.6 mg/kg) and sufentanil (0.2–0.5 μ g/kg) and maintained with remifentanyl in combination with propofol infusion. Patient-controlled intravenous analgesia (PCIA) with sufentanil combined with flurbiprofen was administered for 24–48 h postoperatively as necessary. Intra-abdominal pressure was maintained between 8 and 12 mmHg. Patients received Ringer's lactate solution infusion in the pre- and intraoperative periods at a rate of 10 mL/kg/h [20]. This was continued after surgery until oral fluid intake was allowed.

Outcomes

The primary outcome was the recovery of spontaneous voiding ability after surgery, evaluated by the time between the surgery end point and the first spontaneous voiding time point. The above was recorded in a perioperative bladder diary, a tool commonly recommended for investigating patients with lower LUTS, by the accompanying person [21].

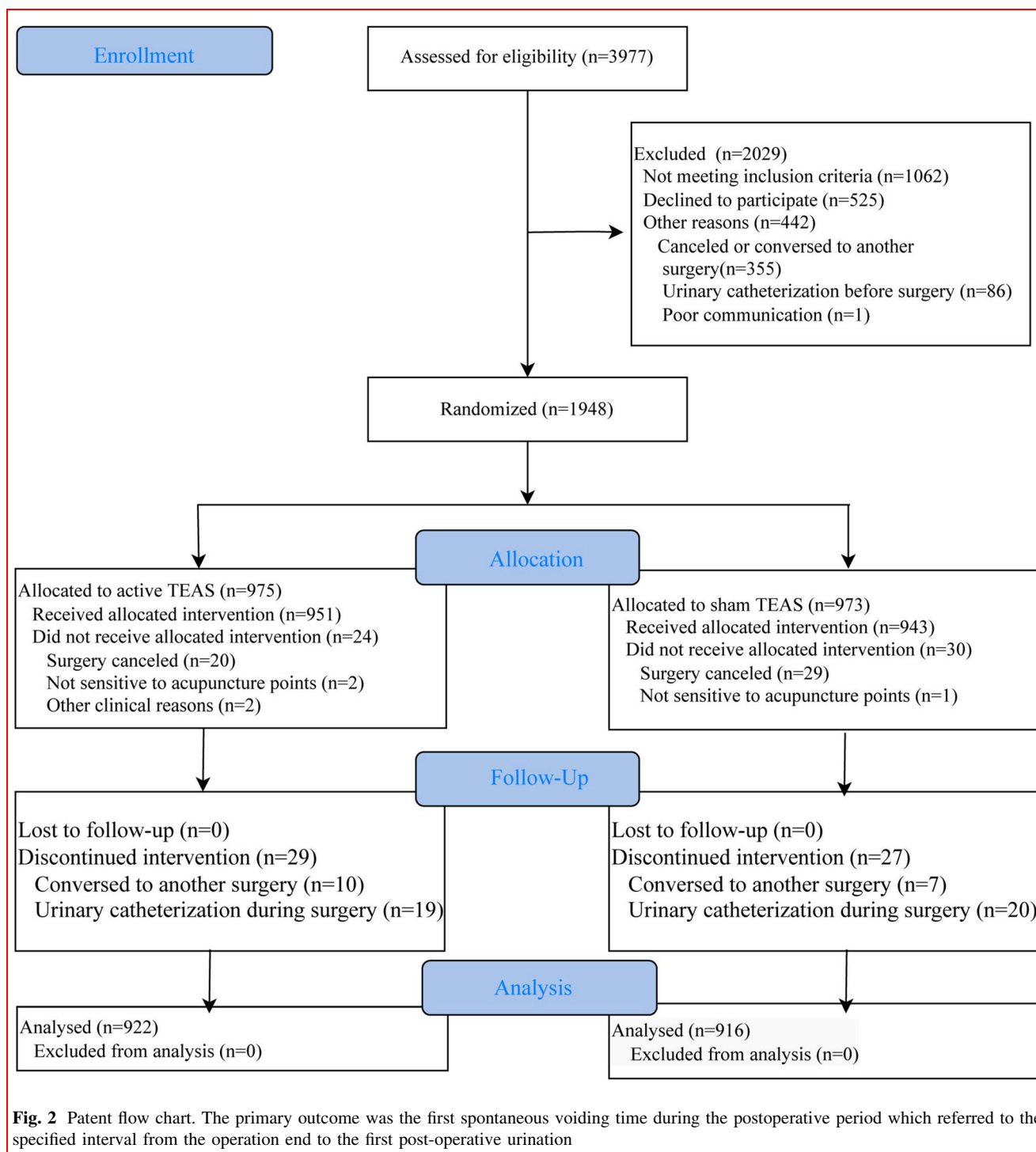
The secondary outcomes included POUR, voiding difficulty, oliguria, nocturia, postoperative pain, sleep quality, anxiety and depression, vomiting, early ambulation, length

of hospital stay, and hospital costs. POUR was diagnosed based on clinical symptoms, ultrasound assessment, and the requirement for bladder catheterization [22]. All treatment-related and treatment-unrelated adverse events were documented throughout the trial.

Statistical analysis

Sample size calculation was based on data from our pilot trial done in 2016 ($n = 50$) in which the delay of recovery of spontaneous voiding ability after surgery was 6.0 ± 3.0 h in the TEAS group and 6.5 ± 3.3 h in the sham group, respectively. The minimum sample size required was 823 patients for each group, at 80% power and one-sided test with a 2.5% significance level. To account for dropout (assuming a dropout rate of approximately 15%), we sought to enroll 970 patients in each group. Statistical analysis was done using PASS software, version 11.0 (NCSS, LLC, Kaysville, UT, USA).

Categorical variables are summarized as frequencies or percentages. The normality of continuous data was tested using the Kolmogorov–Smirnov method ($p < 0.05$: reject normality), and the data are presented as medians with interquartile ranges (IQR) for non-normally distributed data and mean with standard deviation for normally distributed data. Comparisons of the parameters between the TEAS group and the sham group were conducted using the t-test or



Mann–Whitney U test as appropriate. Other binary results were examined using the Chi-squared tests. Kaplan–Meier plots were used to show the time length from the end of surgery to the first time of spontaneous voiding, and p-values were calculated using the log-rank test. The subgroup analyses included: sex, age, operative time, and PCIA. All analyses were conducted using IBM SPSS 22.0 (IBM Corp., Armonk, NY, USA). Statistical significance was set at $p < 0.05$.

Results

Patient characteristics

In total, 3,977 patients were assessed for eligibility for inclusion between August 2018 and June 2020. We excluded 2029 patients for did not satisfy the inclusion criteria, declined to participate, and other reasons. Finally,

1948 patients were eligible for inclusion, and they were randomized to the TEAS ($n = 975$) and sham ($n = 973$) groups. Of these, 53 and 57 patients in the TEAS and sham groups withdrew, respectively. The remaining 1,838 patients completed the study and were included in the analysis. A detailed patient flowchart is shown in (Fig. 2). Patient demographics and baseline characteristics were similar between the groups (Table 1).

Primary outcome

The time length between the ending point of surgery and the first spontaneous voiding time point was 1.3 h (IQR: 0.8–1.8 h) shorter in the TEAS group compared with the sham group, 5.6 h [IQR: 3.7–8.1 h] and 7.1 h [IQR: 4.7–9.7 h], respectively ($p < 0.001$) (Table 2 and Fig. 3). More patients in the sham group underwent delayed urinary recovery (defined as the inability to void urine for > 12 h after induction of anesthesia) compared in the TEAS group (18.9% vs. 11.2%, $p < 0.001$). There was no significant difference in the first spontaneous voided volume between the two groups ($p = 0.72$).

Secondary outcomes and adverse events

A total of 179 patients developed POUR: 75 (8.1%, 95% confidence interval [CI] 6.4–9.9%) and 104 (11.4%, 95% CI 9.3–13.4%) in the TEAS and sham groups, respectively, with statistical difference ($p = 0.02$). After catheter removal, patients in the TEAS group experienced faster spontaneous voiding recovery ($p = 0.027$) and greater volume of first spontaneous voiding ($p = 0.004$) than did those in the sham group (Table 3).

The TEAS group experienced less postoperative voiding difficulty (28.3% vs. 45.1%, $p < 0.001$), less nocturia (23.0% vs. 30.0%, $p < 0.001$), less pain (NRS pain score, 4 [IQR: 3–6] vs. 5 [IQR: 3–6], $p = 0.007$), and less anxiety and depression (HADS score, 0 [IQR: 0–1] vs. 0 [IQR: 0–2], $p < 0.001$) compared with the sham group. There were no significant differences between the two groups in the incidence of postoperative vomiting ($p = 0.38$), length of hospital stay ($p = 0.16$), and hospitalization cost ($p = 0.21$). However, more patients recovered mobilization within 24 h after surgery in the TEAS group than in the sham group (62.1% vs. 56.4%, $p = 0.013$) (Table 4).

No serious adverse events were reported in either group. Only seven cases (0.4%) were reported to have treatment-related or voiding-related adverse events, which were mild and temporary (Table 5).

Table 1 Baseline participant characteristics

Characteristic	TEAS Group ($n = 922$)	Sham Group ($n = 916$)
<i>Sex</i>		
Male	360 (39.0)	383 (41.8)
Female	562 (61.0)	533 (58.2)
<i>Age</i>		
Mean age (y)	58.9 $\pm$ 8.6	59.1 $\pm$ 8.4
Participants aged 40–59	451 (48.9)	457 (49.9)
Participants aged 60–75	471 (51.1)	459 (50.1)
Mean height (cm)	165.2 $\pm$ 7.8	165.5 $\pm$ 7.6
Mean weight (kg)	69.2 $\pm$ 11.8	69.1 $\pm$ 11.3
Mean body mass index (kg/m ²)	25.2 $\pm$ 3.3	25.2 $\pm$ 3.4
<i>Coexisting medical conditions</i>		
Hypertension	359 (38.9)	358 (39.1)
Diabetes mellitus	159 (17.2)	150 (16.4)
Current smoker	159 (17.2)	156 (17.0)
Alcohol use	71 (7.7)	59 (6.4)
Obesity*	56 (6.1)	63 (6.9)
<i>Preoperative preparation</i>		
Enema	421 (45.7)	445 (48.6)
Fasting	64 (6.9)	64 (7.0)
<i>Preoperative micturition</i> [†]		
Median [IQR] urine output (mL)	222.2 (175.0 to 266.7)	216.7 (175.0–266.7)
Median [IQR] voiding interval (h)	6.0 (4.5–7.2)	5.9 (4.4–7.2)
<i>Surgical characteristics</i>		
Median [IQR] operative time (h)	0.8 (0.6–1.1)	0.8 (0.6–1.2)
Median [IQR] intravenous fluid (mL)	750.0 (700.0–800.0)	750.0 (700.0–850.0)
Median [IQR] blood loss (mL)	50.0 (30.0–50.0)	50.0 (30.0–50.0)
Median [IQR] intraoperative opioid dosage, intravenous morphine equivalents (mg) [‡]	30.0 (25.0–35.0)	30.0 (25.0–35.0)
PCIA	837 (90.8)	838 (91.5)
Median [IQR] post-surgery opioid dosage within 48 h, intravenous morphine equivalents (mg) [‡]	110.0 (80.0–110.0)	110.0 (90.0–110.0)

Values are expressed as numbers (percentages) unless otherwise indicated. *Defined as body mass index greater than 30. [†]Preoperative voiding recording started on the afternoon (about 4:00 p.m.) before the operation. [‡]Opioid doses were converted to intravenous morphine equivalents using the ClinCalc.com “Equivalent Opioid Calculator” tool. *BMI* body mass index, *PCIA* patient controlled intravenous analgesia, *IQR* interquartile range

Table 2 The recovery of postoperative spontaneous voiding function

Outcome	TEAS group (n = 922)	Sham group (n = 916)	p value
Median [IQR] time length between the surgery ending point and the first spontaneous voiding time point (h)	5.6 (3.7–8.0)	7.1 (4.7–9.7)	< 0.001
Delayed urinary recovery*	103 (11.2)	173 (18.9)	< 0.001
Median [IQR] volume of the first spontaneous urination(mL)	250 (150.0–380.0)	250 (150.0–400.0)	0.72
Bladder inflation†	93(10.2)	78(8.5)	0.21

Values are expressed as numbers (percentages) unless otherwise indicated. *Defined as inability to void urine for > 12 h after induction of anaesthesia. †Defined as urine volume > 500 mL at first post-operative spontaneous voiding. IQR, interquartile range

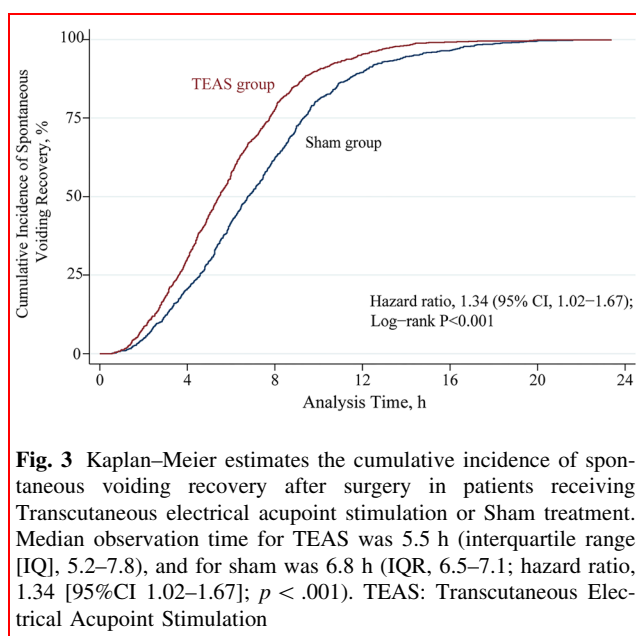


Fig. 3 Kaplan–Meier estimates the cumulative incidence of spontaneous voiding recovery after surgery in patients receiving Transcutaneous electrical acupoint stimulation or Sham treatment. Median observation time for TEAS was 5.5 h (interquartile range [IQ], 5.2–7.8), and for sham was 6.8 h (IQR, 6.5–7.1; hazard ratio, 1.34 [95%CI 1.02–1.67]; $p < .001$). TEAS: Transcutaneous Electrical Acupoint Stimulation

Exploratory analysis

TEAS accelerated the recovery of spontaneous voiding function in both men and women, and TEAS is more effective in men than in women (DeLong test, $p = 0.036$, Fig. 4a). Regardless of surgery duration or age, TEAS showed significant improvement in recovering spontaneous voiding function compared to the sham group (Fig. 4b, c). The DeLong test showed that TEAS was more effective in PCIA than in those without intravenous analgesia ($p = 0.002$, Fig. 4d).

Discussion

In this study, we found that TEAS was an effective treatment for accelerating the recovery of spontaneous voiding after laparoscopic cholecystectomy. It is effective

regardless of sex, age, duration of surgery or PCIA. Moreover, TEAS is more effective in men than in women, as well as in patients with PCIA than those without PCIA. The study also proved that TEAS is effective in relieving pain and anxiety and reducing POUR and voiding difficulty after laparoscopic cholecystectomy.

Spontaneous urination is considered as one of the most important criteria to assess bladder voiding capacity [12, 23]. Moreover, restoring the ability to urinate after surgery is crucial to the prevention and treatment of POUR [24]. Coyle et al. also pointed out that delayed return of normal bladder function can lead to POUR [25]. In addition, the recovery of spontaneous urination within 6 h after surgery was used as a postoperative indicator to guide postoperative catheterization and discharge at the University of Texas Southwestern Medical Center [26]. In our study, the recovery time of postoperative spontaneous urination in patients after TEAS was 5.6 h, which was 1.3 h faster than in the control group and shorter than the threshold of 6 h for catheter insertion mentioned above. A recent study exploring interventions for patients with urinary difficulties after anorectal surgery showed that the median time to spontaneous voiding was 8 h without intervention and 6 h after moxibustion treatment (an acupoint treatment that shares its theoretical background with acupuncture) which was consistent with our study [14].

Several etiologies of postoperative urination disorders exist, including anesthetic disruption of bladder innervation, bladder overdistention, and bladder outlet obstruction [27]. General anesthetics, including sedative-hypnotic and volatile agents, are thought to inhibit autonomic control of urinary function at the level of the pontine micturition center, and voluntary control of the cortex [22]. Urinary retention has adopted many definitions in the literature, and it can best be described as the inability to spontaneously and adequately empty the bladder [22]. However, the diagnosis of POUR is often arbitrary, and its reported incidence widely varies due to the lack of defining criteria. Hakan et al. reported that 4.7% of patients developed

Table 3 Postoperative catheterization

Characteristic	TEAS group (<i>n</i> = 75)	Sham group (<i>n</i> = 104)	<i>p</i> value
Bladder volume (mL)	600.0 (550.0–750.0)	650.0 (612.5–840.0)	0.004
Duration of indwelling catheter (h)	20.5 (15.5–34.0)	21.3 (15.6–32.8)	0.91
Time to spontaneous voiding recovery after catheter removal (h)	0.5 (0.3–1.0)	0.7 (0.4–1.4)	0.027
Volume of first spontaneous voiding after catheter removal (mL)	230.0 (150.0–300.0)	200.0 (150.0–300.0)	0.004

Values are median (interquartile range) unless otherwise stated. *TEAS* transcutaneous electrical acupoint stimulation

Table 4 Symptoms of postoperative spontaneous voiding and early postoperative recovery indicators related to voiding

Outcomes	TEAS group (<i>n</i> = 922)	Sham group (<i>n</i> = 916)	<i>p</i> value
POUR*	75 (8.1)	104 (11.4)	0.02
Voiding difficulty†	261 (28.3)	413 (45.1)	< 0.001
Oliguria‡	20 (2.2)	16 (1.7)	0.51
Nocturia§	212 (23.0)	278 (30.3)	< 0.001
Median [IQR] NRS pain score	4 (3–6)	5 (3–6)	0.007
Median [IQR] NRS sleep disturbance score	3 (2–5)	3 (2–5)	0.053
Median [IQR] HADS score	0 (0–1)	0 (0–2)	< 0.001
Suspected anxiety//	25 (2.7)	38 (4.1)	0.09
Suspected depression//	12 (1.3)	19 (2.1)	0.20
Vomiting	231 (25.1)	246 (26.9)	0.38
Mobilization within 24 h after surgery#	573 (62.1)	517 (56.4)	0.013
Median [IQR] hospitalization duration (d)	7 (6–10)	8 (6–10)	0.16
Median [IQR] hospitalization cost (yuan)	22,714.9 (20,268.5–28,141.5)	23,013.5 (20,395.6–28,916.4)	0.21

Values are expressed as numbers (percentages) unless otherwise indicated. *Diagnosed according to clinical symptoms, ultrasound assessment, and the need for bladder catheterization. †Voiding difficulty referred to the sensation of pain and/or straining to urinate, especially at the beginning of urination. ‡Defined as a urine output < 0.5 mL/kg/h. §Defined as two or more voids per night. //Suspected anxiety and depression were defined as the HADS score greater than 8 for anxiety and depression subscales (range for each subscale, 0–21). # Defined as any partial or full weight-bearing activities conducted 2 to 3 times outside of the ward, with a total walking distance of 250–500 m

NRS numerical rating scale, *HADS* Hospital Anxiety and Depression Scale-Depression, *IQR* interquartile range, *POUR*, postoperative urinary retention

Table 5 Adverse events related to treatment

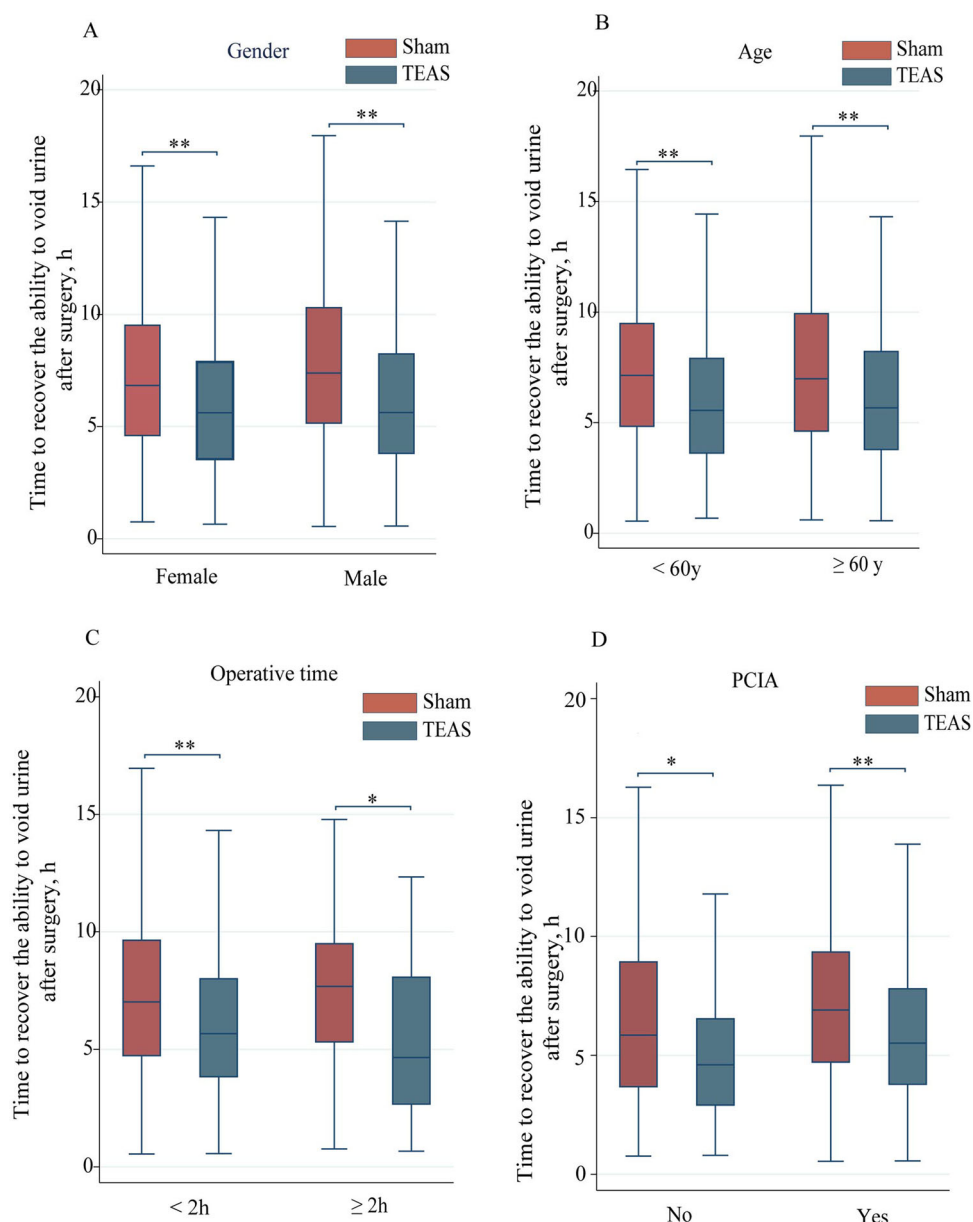
Adverse events	TEAS (<i>n</i> = 922)	Sham TEAS (<i>n</i> = 916)
<i>Voiding-related adverse events</i>		
Gross hematuria	2 (0.1)	1 (0.1)
Lower urinary tract infections	0	0
<i>Treatment-related adverse events</i>		
Redness	2 (0.2)	1 (0.1)
Itching	0	1 (0.1)

Values are expressed as numbers (percentages) unless otherwise indicated

POUR after elective cholecystectomy [6]. In another retrospective cohort conducted by Petros et al., 30% of the patients developed POUR after elective cholecystectomy

[28]. In the present study, we refer to the review in the Journal of Anesthesiology for diagnostic POUR in terms of history and physical examination, need for bladder catheter

Fig. 4 Distribution of times for gender (a), age (b), operative time (c) and PCIA (d) to recover the ability to void urine after surgery. The horizontal line in the box represents the median, upper and lower shaded boundaries show the 25 and 75 quartiles. Any values greater than 1.5 times the interquartile range were considered as outliers for the boxplot, and the box plot excludes these outliers. * $p < 0.05$, ** $p < 0.001$. PCIA: patient-controlled intravenous analgesia



insertion, and ultrasound assessment to make the assessment more accurate and comprehensive [22]. We screened the patients in our study who fulfilled the diagnostic criteria and found the incidence of POUR was 11.4% in patients in the control group, a finding that is consistent with previous results and provides a reference for the incidence of urinary retention after laparoscopic cholecystectomy [4, 5]. The incidence of POUR was significantly reduced to 8.1% after TEAS treatment with statistically significant difference, indicating that TEAS may be an effective approach to reducing the incidence of POUR.

TEAS has been widely used in clinical practice as a noninvasive, economical, and easy treatment for pain,

gastrointestinal disorders, and reproductive medicine [29]. In the present study, the BL32, CV3, and CV4 points were selected as the primary local acupoints and the SP6 points as the distal acupoints. BL-32 is located in the third sacral foramen and plays a role in bladder control by modulating the nervous system [30]. CV3 and CV4 are located near the bladder and have been shown to partially reduce inflammation and edema and regulate vesical pressure by inhibiting hyperpolarization-activated cyclic nucleotide-gated channel protein expression [13, 31]. SP6 is associated with pain management and improved voiding dysfunction when used with BL32, CV3, and CV4; however, the underlying mechanism remains unclear [12, 23, 32].

Strengths and limitations

The strengths of this study include the large sample size. The study also used a sham stimulus to avoid subjective bias in the control group, which made the results more reliable. However, there are several important limitations to this study. First, it was a single-center study, which may have limited generalizability. However, the fact that this study involved 51 surgeons, 25 anesthesiologists, and a relatively large number of patients can mitigate these shortcomings to some extent. Second, despite performed meticulous blinding, there may be a potential placebo effect. Third, we only applied 2-Hz sparse wave signals to the stimulator; other frequencies may be tested and compared in the next step.

In conclusion, this study suggests that perioperative TEAS significantly improves spontaneous urination recovery in patients after laparoscopic cholecystectomy, which may be beneficial for better recovery after surgery. Further studies, such as those conducted in the current trial, are needed to explore the underlying mechanisms of TEAS.

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Declarations

Conflict of interest No authors have any conflicts of interest or funding to disclose at this time.

Ethical approval The study was approved by the appropriate institution and national research ethics committee.

Informed consent Informed, written consent was obtained from all individual participants included in the study.

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